Destination-Aware Coordinated Ramp Metering for Preventing Off-Ramp Queue Spillover and Mainstream Congestion



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***Abstract*—**Freeway congestion usually occurs in on-ramp areas; however, it also occurs around off-ramp areas and can even spill back onto the main line, which is common on many congested urban freeways during peak hours. Nevertheless, traditional ramp metering strategies, one of the most successful approaches to reducing freeway congestion, have not considered the off-ramp traffic dynamics. Therefore, this study proposes a destination-aware coordinated ramp metering model for preventing off-ramp queue spillover and mainstream congestion. The model is established based on a model predictive control, which contains a traffic evolution module and an optimization module. The traffic evolution module is described in a destination-aware manner with explicit consideration of the impact on mainstream traffic due to off-ramp queue spillover. The optimization module especially takes into account the off-ramp queue constraints so as to prevent the spillover. A modified feasible direction algorithm is designed for solving the optimization problem. A comprehensive case study is conducted on a congested urban freeway network in Shanghai, China, during peak hours. A sensitivity analysis demonstrates its capabilities and robustness in preventing off-ramp queue spillover and freeway congestion. The results show that the proposed strategy reduces the total time spent by 8.3% with respect to the no-control case, outperforming the traditional ramp metering strategies without consideration of off-ramp traffic dynamics.**

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Urban freeways are major infrastructures for enabling the efficient movement of people and goods on road networks, and traffic congestion has long been a major problem on many freeways worldwide [1], [2], [3]. One of the most widespread measures for reducing freeway congestion is ramp metering, which regulates the freeway inflows at on-ramps using traffic signals. It mainly aims to resolve congestion on the freeway main line and, especially, in on-ramp merging areas [4].

Introduction

Besides the on-ramp areas, congestion also occurs around off-ramp areas owing to the limited capacities of downstream at-grade intersections, as illustrated in Figure 1. The downstream capacities of the off-ramps are limited by inappropriate signal control at the intersections and/or weaving effects on the surface roads. Such bottlenecks result in an off-ramp queue and congestion spillback onto the main line, greatly reducing the operational efficiency (capacity) of the urban freeway [5], [6], [7]. This phenomenon is more serious when the distance between the off-ramp and downstream intersection is short. The core reason is that the short-term off-ramp volumes exceed the downstream capacities. Additionally, it can be seen from Figure 1(b) that the distance between the off-ramp and its upstream on-ramp is also quite short. Therefore, the limited capacities downstream of the off-ramps should be considered by on-ramp metering strategies to prevent spillover. A destination-aware coordinated ramp metering (CRM) is proposed in this study aiming not only to avoid mainstream inflows exceeding the main-line capacity but also to adjust the mainstream outflow at off-ramps according to their downstream capacities. The mainstream inflows are regulated at on-ramps to influence the mainstream outflows at downstream off-ramps.

Related Work and Motivation

The problem of off-ramp queue spillover has mostly been discussed at local levels (limited to off-ramp areas). To mitigate the weaving effects at congested surface streets, Spiliopoulou et al. [8] regulated off-ramp exiting flow via a signal control device located at the off-ramp terminal with the aim of maximizing the outflow of the weaving bottleneck. Zhao and Liu [9] jointly optimized lane assignment and signal control of the intersection downstream of the off-ramp. A presignal strategy and sorting area have been further introduced to increase the capacity of the downstream intersection [10].

On the basis of off-ramp queue detection and prediction, Yang et al. [11] integrated adaptive signal control and off-ramp priority control to prevent off-ramp queue spillover. Liu et al. [12] optimized off-ramp terminal location to improve traffic efficiency and safety in the weaving areas at the surface streets. Zheng et al. [13] proposed a cooperative lane-changing strategy for connected and automated vehicles to enable smooth diverging near freeway off-ramps. On the basis of modeling freeway traffic congestion triggered by off-ramp queue spillover, Chen et al. [14]

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proposed a signal optimization with explicit consideration of lane-changing maneuvers at freeway diverging areas. The aforementioned strategies increased the downstream capacities of the off-ramps; nevertheless, queue spillover could not be prevented consistently if the off-ramp traffic demand was too intense. Hence, it is equally necessary to regulate the mainstream outflow at off-ramps via appropriate control strategies.

A few strategies have been proposed from networkwide perspectives for regulating off-ramp volumes and preventing queue spillover. Considering simplified road networks, Haddad et al. [15] controlled freeway and urban networks in an integrated manner to avoid spillback. Spiliopoulou et al. [16] proposed route diversion, i.e., rerouting vehicles from congested off-ramp areas to unsaturated nearby off-ramps toward the same destination. In contrast to these works, which mainly focused on the optimization of large-scale but simplified road networks, this study pays special attention to traffic dynamics among freeway sections and optimizes the traffic operation of freeway networks consisting of main lines, on-ramps, and off-ramps. Ramp metering, as one of the most typical and effective freeway control measures, is specifically investigated to regulate mainstream inflows and affect mainstream outflows for preventing congestion and queue spillover.

Various ramp metering strategies have been developed since the 1960s. Wattleworth [17] first proposed a fixed-time ramp metering strategy to avoid mainstream inflows exceeding capacity. Since the 1980s, with real-time measurements becoming more widespread, many reactive strategies have been proposed and practiced, including local ramp metering (LRM) and CRM.

LRM strategies mainly aim to prevent the onset of congestion in merging areas, and the metering rates for each on-ramp are calculated independently. One of the most commonly used LRM strategies is Asservissement Linéaire d'Entrée Autoroutière (ALINEA), a local feedback-based traffic controller [18]. Building on top of ALINEA, various variants have been proposed [19], [20], [21]. These LRM strategies are easy to implement to address local congestion, but they may not bring improvements at network levels. For example, reducing upstream congestion enables vehicles to run faster toward the downstream but may lead to downstream congestion and off-ramp queue spillover. Therefore, CRM strategies should be proposed to optimize the entire network.

CRM strategies simultaneously control a set of ramps to enhance the traffic efficiency of a freeway network in a systematic manner. They can be classified into two groups: rule-based and optimization-based strategies. Many CRM strategies in practice are rule based, e.g., BOTTLENECK [22], SWARM [23], and HERO [24]. Though easy to implement, the rule-based control will not activate until the off-ramp queue spills back to the mainline detectors. Even if the rule-based control activates in advance of a spillover, it is not guaranteed to achieve the optimal control effect.

The optimal control effect can be theoretically guaranteed through optimization-based strategies, and

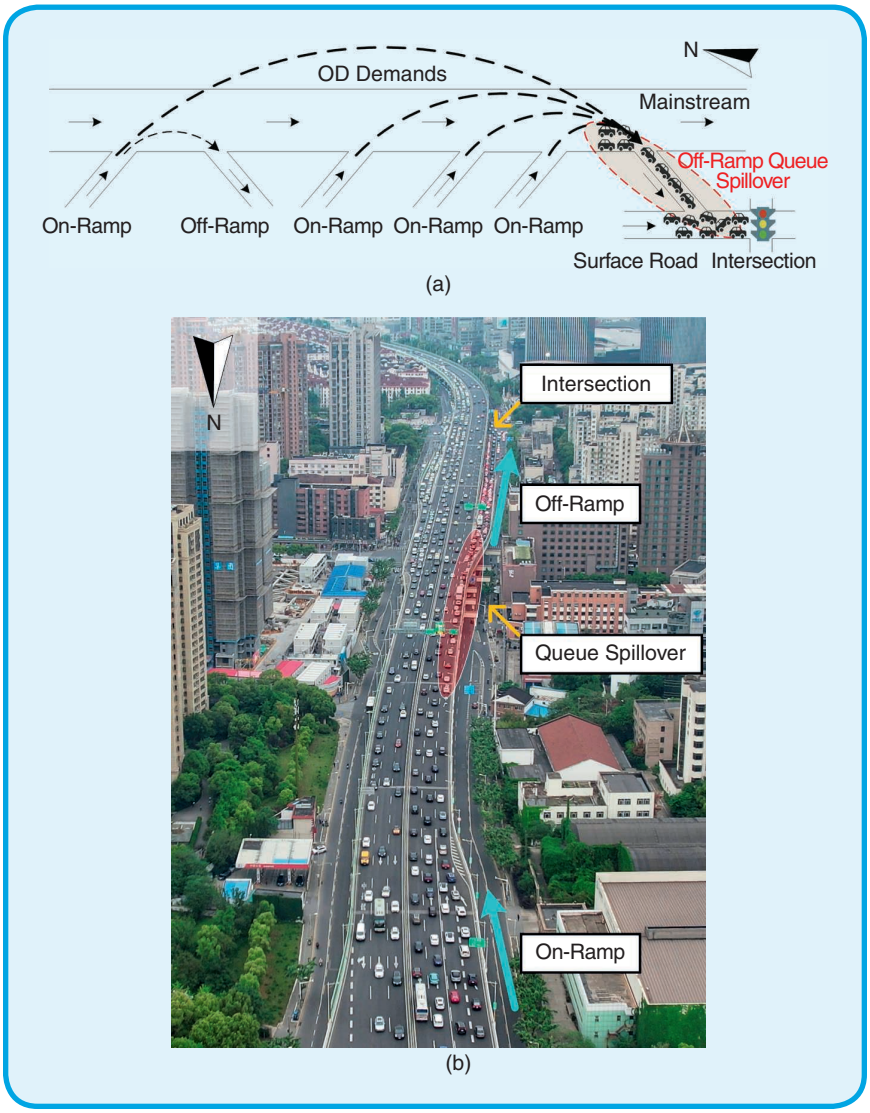


FIG 1 The off-ramp queue spillover in urban freeway traffic systems: (a) a sketch and (b) a site photo.

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such problems can be formulated as optimal control problems. Some optimization-based strategies control actions over a finite horizon, such as the well-known "Advanced Motorway Optimal Control" [25]. Other strategies optimize over a prediction horizon and update the optimization through a rolling horizon. The optimal control problem is solved repeatedly in real time to manage the accumulated errors.

The optimization was typically embedded in model predictive control (MPC) algorithms, in which traffic models have been extensively adopted to describe the traffic flow dynamics along freeway main lines [26], [27], [28], [29], [30], [31], [32]. To calculate the off-ramp traffic volumes exiting from the freeway main line, most studies have introduced preset off-ramp split ratios (also called *turning rates*), either fixed [32], [33] or time dependent [34]. The objectives of the optimal control problems are usually to minimize the total time spent (TTS) of the studied network, typically including a main line and on-ramps [29], [33], [34]. Regarding the constraints, the mainstream and on-ramp flows are upper-bounded by the road capacity. The on-ramp storage space has also been widely considered in constraints aiming to prevent on-ramp queue spillovers onto surface streets when the metering rates are small for a long period of time. However, the equally important off-ramp queue constraints have not received comparable attention in traditional ramp metering strategies. Consequently, even if optimization-based ramp metering strategies manage to resolve congestion on main lines and on-ramps, without consideration of the limited off-ramp capacity, off-ramp queue spillover may still occur.

In summary, the LRM and rule-based CRM strategies can hardly achieve the optimal control actions at network levels. The optimization-based CRM strategies so far have mainly considered only on-ramps and the freeway main line. Consequently, the "optimal" control actions may still result in off-ramp queue spillover. In contrast to traditional ramp metering strategies, this study explicitly considers off-ramp queue dynamics in the optimization.

Contributions and Article Organization

The main contributions of this article are summarized as follows:

- The concept of ramp metering strategies is extended to prevent not only mainstream congestion but also off-ramp queue spillover. Vehicles are not allowed to enter freeways if they will cause potential off-ramp queue spillbacks downstream.
- A destination-aware ramp metering strategy is proposed. It is formulated as an optimal control problem with explicit consideration of off-ramp queue constraints and the impact on mainstream traffic due to off-ramp queue spillover. The optimization is embedded in an MPC algorithm with a rolling horizon to enable dynamic traffic control.
- A comprehensive analysis is conducted to demonstrate the effectiveness, necessity, and robustness of the proposed ramp metering strategy regarding to preventing off-ramp queue spillover and mainstream congestion.

The rest of this article is organized as follows. The "Methodology" section formulates the optimal control problem for a freeway network comprising a main line, on-ramps, and off-ramps and then designs a modified feasible direction algorithm for solving the problem. The "Case Study" section presents the results from the case studies and compares the proposed control strategy and solution algorithm against traditional ones. The "Sensitivity Analysis" section conducts a sensitivity analysis regarding the noise origin–destination information and different control parameter settings. Finally, the "Conclusions" section concludes this article and recommends potential further research directions.

Methodology

This section describes destination-aware CRM for preventing off-ramp queue spillover and mainstream congestion. Traffic variables, such as the traffic flow and density, are distinguished by destinations. The destination-aware traffic models are discrete in space and time. The freeway stretch is divided into main-line sections indexed with a, the traffic variables are distinguished by destination j, and the time horizon is discretized into time intervals indexed with t.

MPC Formulation

MPC has been used extensively and successfully for optimal traffic control with constraints [35], [36], [37], [38]. In this study, the control problem is formulated as an optimal control problem, and the optimization is embedded in an MPC algorithm with a rolling horizon, as depicted in Figure 2. It predicts the evolution of the traffic flow, evaluates the control performance, and optimizes control actions over a prediction horizon N_p. The traffic variables (including the flow, density, and vehicle queue) are expressed in a destination-aware mode and serve as the inputs to the controller in Figure 2. Inside the controller, the traffic evolution module and optimization module are conducted to search for the optimal control action, i.e., the optimal metering rates, for optimizing the control performance over the prediction horizon. To adapt to a highly dynamic traffic system, the control strategy is updated with each control step size $N_c (\leq N_p)$. In other words, the control strategy only executes the first N_c time intervals of the calculated control actions.

Input: Traffic States

The state of the urban freeway traffic system $\mathbf{x}_t$ contains the traffic states of the main-line sections, on-ramps, and off-ramps at the beginning of time interval t, denoted by $\mathbf{x}_t^{\text{MAIN}}$, $\mathbf{x}_t^{\text{ON}}$, and $\mathbf{x}_t^{\text{OFF}}$, respectively. They are given by $\mathbf{x}_t = \mathbf{x}_{1,t}^{\text{MAIN}}, \mathbf{x}_{2,t}^{\text{MAIN}}, \ldots, \mathbf{x}_{a,t}^{\text{MAIN}}, \ldots, \mathbf{x}_{1,t}^{\text{ON}}, \mathbf{x}_{2,t}^{\text{ON}}, \ldots, \mathbf{x}_{i,t}^{\text{ON}}, \ldots,$

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$\mathbf{x}_{1,t}^{\text{OFF}}$, $\mathbf{x}_{2,t}^{\text{OFF}}$, …, $\mathbf{x}_{j,t}^{\text{OFF}}$, …), in which $\mathbf{x}_{a,t}^{\text{MAIN}} = (\rho_{a,j,t}, q_{a,j,t}, v_{a,t})$ for main-line section $a \in \boldsymbol{A}$, $\mathbf{x}_{i,t}^{\text{ON}} = (d_{i,j,t}, N_{i,j,t}^{\text{ON}}, r_{i,j,t})$ for on-ramp $i \in \boldsymbol{I}_{\text{ramp}}$ and $x_{j,t}^{\text{OFF}} = (e_{j,t}, N_{j,t}^{\text{OFF}})$ for off-ramp $j \in \boldsymbol{J}_{\text{ramp}}$. Here, $\rho_{a,j,t}$ and $q_{a,j,t}$ are traffic density and traffic flow heading to destination j in main-line section a, respectively. $v_{a,t}$ is the mean traffic speed in main-line section a during the last time interval. $d_{i,j,t}$, $N_{i,j,t}^{\text{ON}}$, and $r_{i,j,t}$ represent the demand flow, number of queueing vehicles, and exiting flow heading to destination j at on-ramp i, respectively. $e_{j,t}$ and $N_{j,t}^{\text{OFF}}$ are, respectively, the entering flow and number of vehicles queueing at off-ramp j. By aggregating $\rho_{a,j,t}$, $q_{a,j,t}$, $d_{i,j,t}$, $N_{i,j,t}^{\text{ON}}$, and $r_{i,j,t}$ with all destinations, we can obtain the total quantities of these traffic states in a nondestination-aware mode, i.e., $\rho_{a,t}$, $q_{a,t}$, $d_{i,t}$, $N_{i,t}^{\text{ON}}$, and $r_{i,t}$.

Output: Control Action

The control action $\mathbf{u}_t$ for this problem comprises the metering rates of the controllable on-ramps, as given by $\mathbf{u}_t = (\mu_{1,t}, \mu_{2,t}, \ldots, \mu_{i,t}, \ldots)$, where $\mu_{i,t} \in [\mu_i^{\min}, 1]$ is the metering rate of on-ramp i. If $\mu_{i,t} = 1$, no ramp metering control is implemented. A smaller value of $\mu_{i,t}$ means a lower metering rate and stricter limitation of the traffic flow. A minimum metering rate $\mu_i^{\min}$ is prespecified to prevent infinite waiting times at on-ramps. The control actions are calculated over the entire prediction horizon, but only the first one is applied. The optimization is updated with a rolling horizon [39], [40].

Traffic Evolution Module

The traffic dynamics are formulated as macroscopic traffic models. The traffic evolution is described from the upstream on-ramp to the main line and, eventually, to the downstream off-ramp.

For on-ramp i within main-line section a, the entering flow $r_{i,t}$ is influenced by the control action of metering rate $\mu_{i,t}$. It can be determined as follows:

$$r_{i,t} = \min\left\{ d_{i,t} + \frac{N_{i,t}^{\text{ON}}}{T}, \mu_{i,t} r_i^{\max}, r_i^{\max} \frac{\rho_a^{\text{J}} - \rho_{a,t}}{\rho_a^{\text{J}} - \rho_a^{\text{C}}} \right\} \tag{1}$$

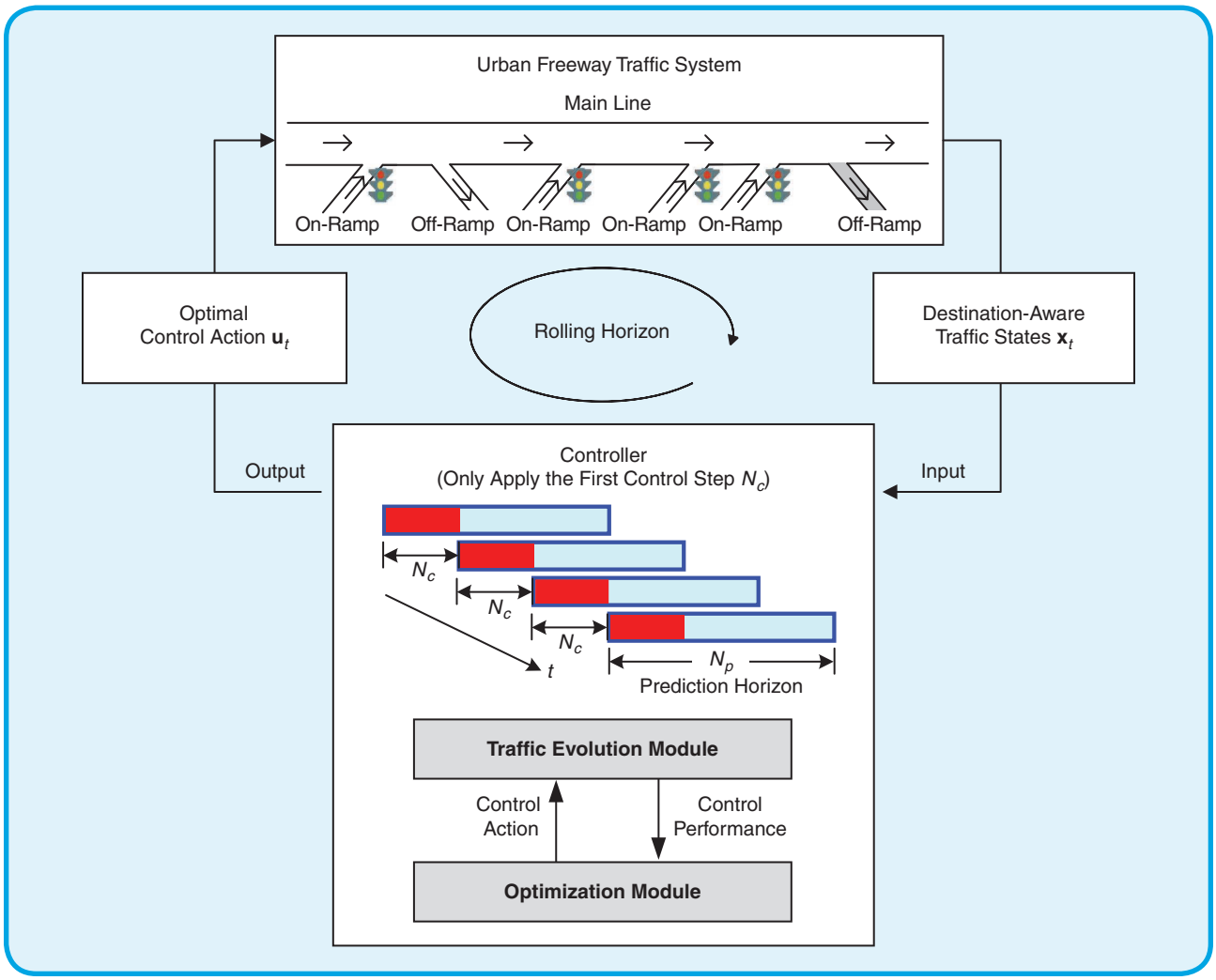


FIG 2 The framework of the proposed destination-aware CRM.

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where $d_{i,t}$ and $N_{i,t}^{\mathrm{ON}}$ represent the demand flow and number of queueing vehicles at on-ramp i during time interval t; T is the length of one time interval; $r_i^{\max}$ is the capacity flow of on-ramp i; $\rho_{a,t}$ is the traffic density in main-line section a at the end of time interval t; and ρ_a^{J} and ρ_a^{C} are the jam density and critical density of main-line section a, respectively.

The on-ramp queue dynamics can then be described in a destination-aware manner, as follows:

$$N_{i,j,t+1}^{\mathrm{ON}} = N_{i,j,t}^{\mathrm{ON}} + (d_{i,j,t} - r_{i,j,t})T \tag{2}$$

Here, $r_{i,j,t}$ is the entering flow heading to destination j at on-ramp i. It can be calculated as $r_{i,t}\gamma_{i,j,t}$, where $\gamma_{i,j,t}$ is the composition rate, and there always exists $\Sigma_{j\in\mathcal{J}}\gamma_{i,j,t} = 1$.

For main-line section $a \in \boldsymbol{A}$, the traffic evolution is given by the following equations; (3) updates the traffic density $\rho_{a,j,t+1}$ according to the conservation law:

$$\rho_{a,j,t+1} = \rho_{a,j,t} + \frac{T}{L_a\lambda_a}(q_{a-1,j,t} - q_{a,j,t} + r_{i,j,t}) \tag{3}$$

where L_a and λ_a denote the length and number of lanes of main-line section a.

The lane-drop impact on mainstream speed caused by an off-ramp queue spillover is explicitly considered. The speed evolution of $v_{a,t+1}$ is described by a second-order macroscopic traffic model, as given by (4). Given an off-ramp j located within the section $(a+1)$, a binary variable $\phi_{a+1,t}$ is introduced in this study to indicate whether the off-ramp queue $N_{j,t}^{\mathrm{OFF}}$ is larger than its storage capacity S_j^{OFF}. If yes, that will create the blockage of the rightmost lane on the main line (as illustrated by Figure 3), and, thus, $\phi_{a+1,t} = 1$, and the lane-drop impact term is activated; otherwise, $\phi_{a+1,t} = 0$:

$$\begin{aligned} v_{a,t+1} = v_{a,t} &+ \frac{T}{\tau}[V(\rho_{a,t}) - v_{a,t}]\,(\text{relaxation term}) \\ &+ \frac{T}{L_a}v_{a,t}(v_{a-1,t} - v_{a,t})\,(\text{convection term}) \\ &- \frac{\nu T(\rho_{a+1,t} - \rho_{a,t})}{\tau L_a(\rho_{a,t} + \kappa)}\,(\text{anticipation term}) \\ &- \frac{\delta T r_{i,t} v_{a,t}}{L_a\lambda_a(\rho_{a,t} + \kappa)}\,(\text{on-ramp impact term}) \\ &- \phi_{a+1,t}\frac{\varphi T\Delta\lambda\rho_{a,t}v_{a,t}^2}{L_a\lambda_a\rho_a^{\mathrm{C}}}\,(\text{lane-drop impact term}) \end{aligned} \tag{4}$$

where τ, ν, κ, δ, and φ are parameters regarding driver behaviors [7], and $\Delta\lambda$ is the number of dropped lanes caused by off-ramp queue spillover.

The impact on the mainstream fundamental diagram due to an off-ramp queue spillover is also taken into account. According to an empirical study [41], incidents with lane closures bring reductions on free flow speed, critical density, and capacity flow per lane. As depicted in Figure 3, if an off-ramp queue spill backs up to main-line section a (i.e., $\phi_{a,t} = 1$), the free flow speed v_a^{F} drops to $v_a^{\mathrm{F}}(1 - \phi_{a,t}\xi_v)$, and the critical density ρ_a^{C} decreases to $\rho_a^{\mathrm{C}}(1 - \phi_{a,t}\xi_\rho)$, where ξ_v and ξ_ρ are parameters representing the impact of off-ramp queue spillover. On the basis of the adjusted fundamental diagram, the steady-state speed $V(\rho_{a,t})$ in (4) can be calculated using (5), where the constant parameter α_a reflects the traffic flow characteristics of main-line section a:

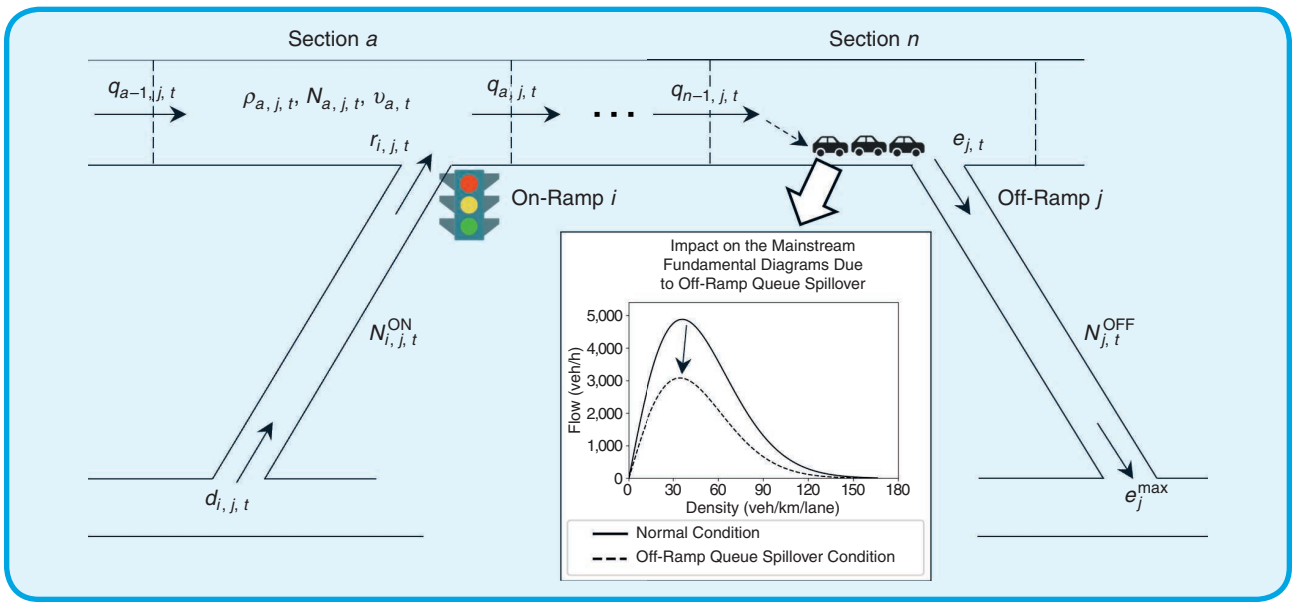


FIG 3 The formulation of destination-aware traffic dynamics with possible off-ramp queue spillovers. veh: vehicles.

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$$V(\rho_{a,t}) = v_a^{\mathrm{F}}(1-\phi_{a,t}\xi_v)\cdot\exp\left[-\frac{1}{\alpha_a}\left(\frac{\rho_{a,t}}{\rho_a^{\mathrm{C}}(1-\phi_{a,t}\xi_\rho)}\right)^{\alpha_a}\right]. \quad (5)$$

Equation (6) calculates destination-aware mainstream traffic flows $q_{a,j,t+1}$ by updated traffic flow $q_{a,t+1}$ and corresponding composition rates $\vartheta_{a,j,t+1}$. The traffic flow $q_{a,t+1}$ is predicted by multiplying the updated density $\rho_{a,t+1}$, speed $v_{a,t+1}$, and lane number $(\lambda_a - \phi_{a,t}\Delta\lambda)$:

$$\begin{aligned} q_{a,j,t+1} &= q_{a,t+1}\vartheta_{a,j,t+1} \\ &= \rho_{a,t+1}(\lambda_a - \phi_{a,t}\Delta\lambda)\, v_{a,t+1}\vartheta_{a,j,t+1}. \end{aligned} \quad (6)$$

The off-ramp traffic dynamics are specifically formulated in this study. As illustrated by Figure 3, for an off-ramp j located in main-line section n $(n > a)$, the off-ramp demand flow $q_{n-1,j,t+1}$ can be described as $q_{n-1,t+1}\cdot\vartheta_{n-1,j,t+1}$ according to (6), where $q_{n-1,t+1}$ is the total traffic flow of section $(n-1)$ located at the immediate upstream of off-ramp j. Particularly, $\vartheta_{n-1,j,t+1}$ is the composition rate regarding the destination of off-ramp j, and it is dynamically updated according to (1)–(5) rather than preset split ratios.

The mainstream outflow $e_{j,t+1}$ at off-ramp j can be determined by the demand flow of the exiting traffic and capacity of off-ramp j to receive the demand flow. The equation is given as follows:

$$e_{j,t+1} = \min\left\{q_{n-1,j,t+1}, \frac{S_j^{\mathrm{OFF}} - N_{j,t}^{\mathrm{OFF}}}{T}\right\} \quad (7)$$

According to the conservation law, the off-ramp queue $N_{j,t+1}^{\mathrm{OFF}}$ of the next time interval $(t+1)$ can be predicted as follows:

$$N_{j,t+1}^{\mathrm{OFF}} = \max\left\{N_{j,t}^{\mathrm{OFF}} + (e_{j,t+1} - e_j^{\max})T, 0\right\} \quad (8)$$

Here, $e_j^{\max}$ is the capacity at the downstream of off-ramp j.

Optimization Module

The optimization module embedded in the described MPC algorithm is formulated as follows.

Constraints

To prevent on- and off-ramp queue spillover, constraints (9a) and (9b) ensure that the on-ramp queue $N_{i,t}^{\mathrm{ON}}$ and off-ramp queue $N_{j,t}^{\mathrm{OFF}}$ do not exceed their corresponding storage capacities S_i^{ON} and S_j^{OFF}. The effect of the off-ramp queue constraints is analyzed in the "Effectiveness of the Proposed Control Model" section in the case study:

$$N_{i,t}^{\mathrm{ON}} \le S_i^{\mathrm{ON}}, \quad \forall i \in \boldsymbol{I}_{\mathrm{ramp}}, \quad t \in [t_0, \ldots, t_0 + N_{\mathrm{p}} - 1] \quad (9a)$$

$$N_{j,t}^{\mathrm{OFF}} \le S_j^{\mathrm{OFF}}, \quad \forall j \in \boldsymbol{J}_{\mathrm{ramp}}, \quad t \in [t_0, \ldots, t_0 + N_{\mathrm{p}} - 1]. \quad (9b)$$

The optimization is also subject to the constraints of the traffic models, which have been introduced in the traffic evolution module.

Objective Functions

The objective is to minimize the TTS of the entire freeway network over the prediction horizon $[t_0, \ldots, t_0 + N_{\mathrm{p}} - 1]$, where t_0 denotes the current time interval, and N_{p} indicates the length of the prediction horizon. The TTS includes not only the total travel time (TTT) on the mainstream and total waiting time (TWT) at on-ramps (denoted by $\mathrm{TWT}^{\mathrm{ON}}$) but also the TWT at off-ramps (denoted by $\mathrm{TWT}^{\mathrm{OFF}}$). This is to ensure that the off-ramp traffic dynamics are considered in the optimization. In addition, constraints (9a) and (9b) are considered as the last two penalty terms in the objective function with a sufficiently large weight M (e.g., 1,000). Therefore, the objective function can be written as

$$\begin{aligned} \min J = &\sum_{t=t_0}^{t_0+N_{\mathrm{p}}-1}\sum_{a\in\boldsymbol{A}} N_{a,t}T + \sum_{t=t_0}^{t_0+N_{\mathrm{p}}-1}\sum_{i\in\boldsymbol{I}_{\mathrm{ramp}}} N_{i,t}^{\mathrm{ON}}T + \sum_{t=t_0}^{t_0+N_{\mathrm{p}}-1}\sum_{j\in\boldsymbol{J}_{\mathrm{ramp}}} N_{j,t}^{\mathrm{OFF}}T \\ &+ M\cdot\sum_{t=t_0}^{t_0+N_{\mathrm{p}}-1}\sum_{i\in\boldsymbol{I}_{\mathrm{ramp}}}\max\left\{N_{i,t}^{\mathrm{ON}} - S_i^{\mathrm{ON}}, 0\right\} \\ &+ M\cdot\sum_{t=t_0}^{t_0+N_{\mathrm{p}}-1}\sum_{j\in\boldsymbol{J}_{\mathrm{ramp}}}\max\left\{N_{j,t}^{\mathrm{OFF}} - S_j^{\mathrm{OFF}}, 0\right\}. \end{aligned} \quad (10)$$

Solution Algorithm

The aforementioned optimal control problem can be solved by a feasible direction algorithm incorporating a derivative backpropagation method, i.e., "resilient backpropagation" (RPROP) [42]. The RPROP is a gradient-based iteration process aiming to efficiently converge to the optimal control actions. It has been adopted to solve large-scale nonlinear optimization problems in the field of freeway traffic control, and its effectiveness has been demonstrated [25], [33].

However, the regular RPROP algorithm sometimes converges to a local minimum, as freeway traffic optimal control problems are often nonlinear and nonconvex [43]. Therefore, a modified RPROP algorithm is designed with a stopping criterion to not only examine whether the results are convergent but also to verify whether the convergence value is close to the minimum of each iteration. If both are satisfied, terminate the iteration process; otherwise, return to the control action corresponding to the minimum and search within its neighborhood with smaller iteration steps (decelerate the convergence in this region) in case the algorithm jumps over the minimum. The flowchart of the modified RPROP algorithm is demonstrated by Figure 4. Note that the modified RPROP does not guarantee avoiding a local minimum, but the control action $\mathbf{u}_t$ optimized by the modified RPROP is no worse than the one obtained by the regular RPROP. The benefits of the modified RPROP algorithm are presented in the "Effectiveness of the Modified Solution Algorithm" section.

Case Study

A comprehensive case study was conducted to verify the effectiveness of the proposed strategy. The following sections first introduce the background of the case study, then

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compare the control performances of different strategies, and finally compare the iteration processes of the modified RPROP algorithm against the regular RPROP.

Background

A three-lane urban freeway network in Shanghai, China, where on- and off-ramps are densely located is considered for the case study. There are five on-ramps and three off-ramps within the selected stretch. The freeway stretch is 5.4 km in length, and it was divided into 18 main-line sections, as shown in Figure 5. The parameters of the freeway network were determined according to real traffic systems (see "Network Parameters").

The selected stretch is one of the most congested freeways in Shanghai, China. Its morning peak usually begins at approximately 6:40 a.m. This case study simulated the traffic dynamics of a typical morning peak. The simulation period was from 5 to 9 a.m. A dataset of the traffic demand was prepared for the simulation inputs, including each vehicle's origin, destination, and arrival time at the origin. The arriving demands at each origin were the same as the traffic counts for 5 min according to real loop detector data on a typical weekday (25 October 2017, a Wednesday), as shown in Figure 6. The destination distributions of the arriving demands at each origin for 5 min were aligned with the origin-destination (OD) distributions as extracted from

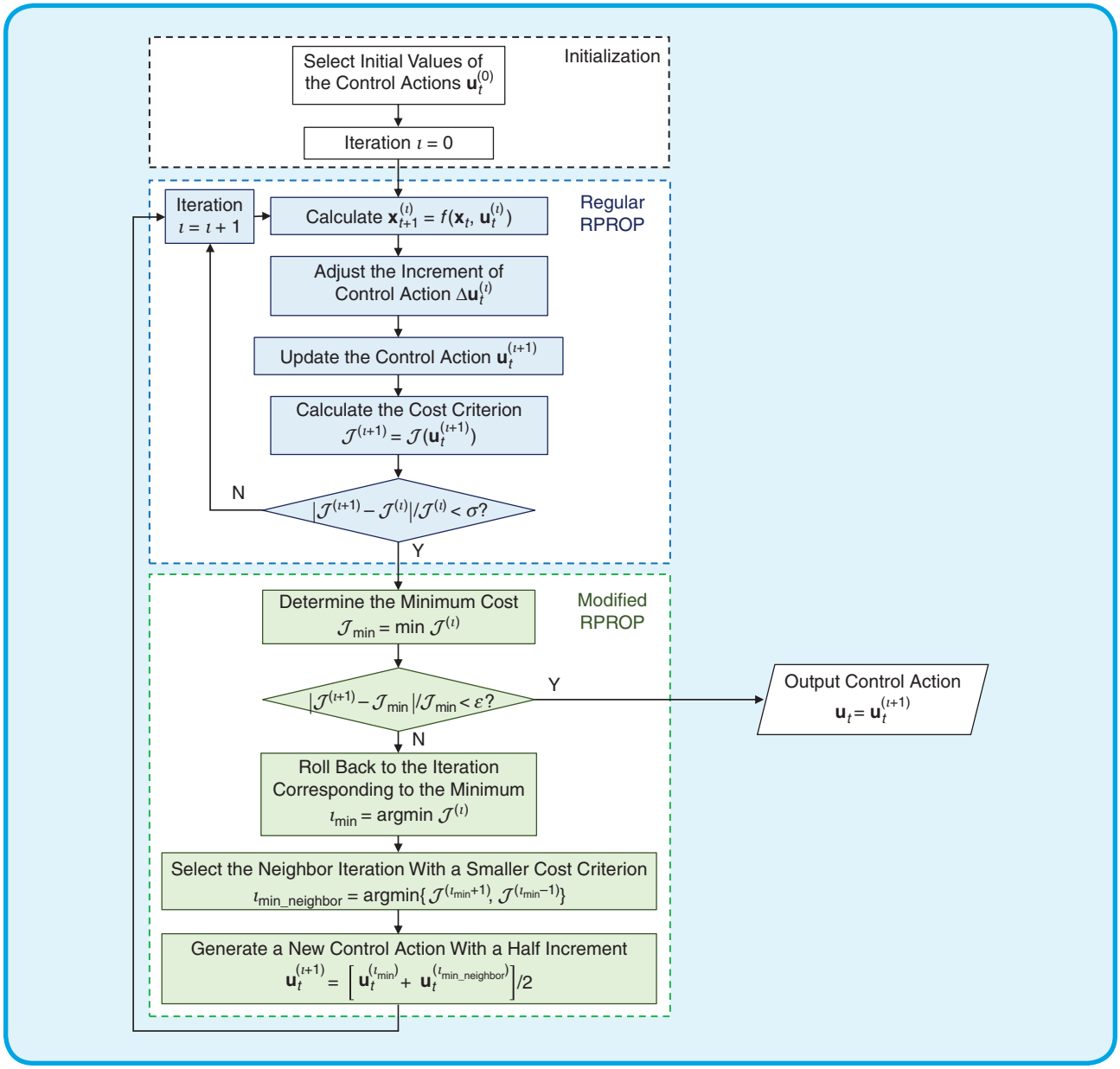


FIG 4 A flowchart of the modified RPROP algorithm.

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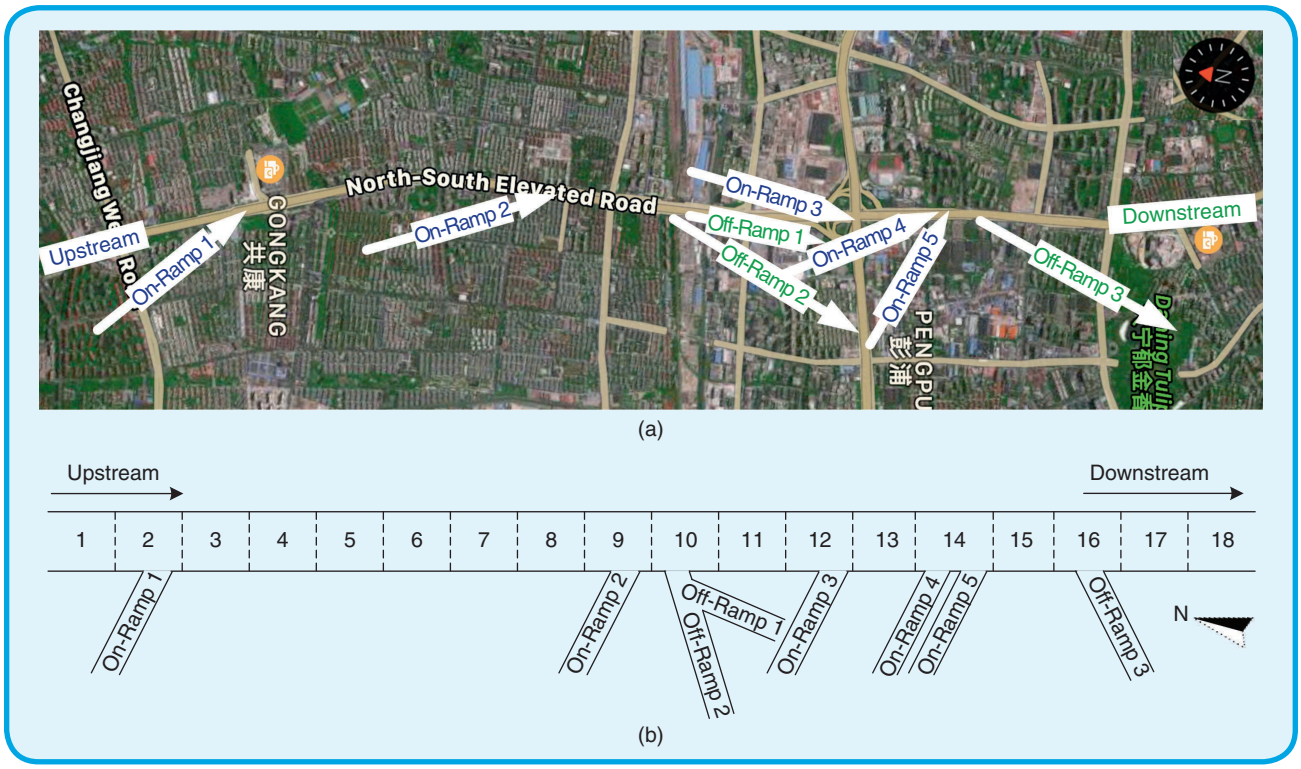


FIG 5 The urban freeway network of a north–south elevated road in Shanghai, China, for the case study: (a) a satellite map and (b) the discretization of the network. N: north.

real vehicle trajectory data (provided by Amap—a mobile navigation application with high market penetration in China). Vehicle trajectory data for the same simulation period on five weekdays (5–9 a.m., 23–27 October 2017) were utilized to obtain the OD distributions of the studied freeway network. The representativeness of vehicle trajectory data and their resulting OD samples could be verified using loop detector data by the work of [44].

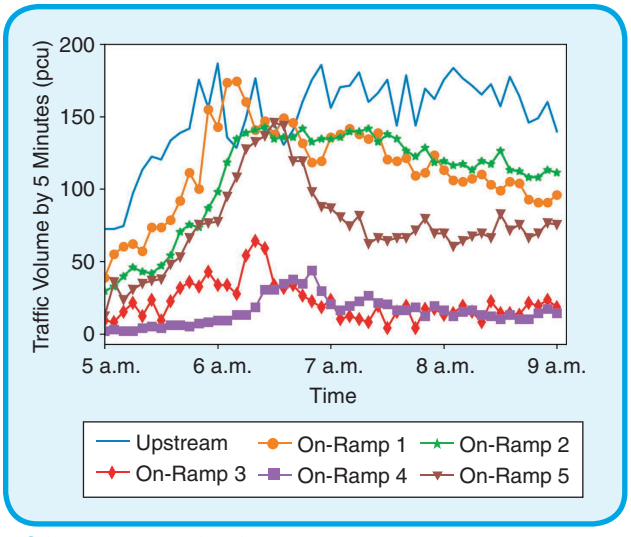


FIG 6 The demand profile of the simulation. pcu: passenger car unit.

The traffic flow evolution was simulated by macroscopic traffic models, and we selected $T = 10$ s as one simulation time interval. The following parameters were used in (4): $\tau = 36$ s, $\nu = 30$ km^2/h, $\kappa = 40$ vehicles/km/lane, $\delta = 0.33$ h/km, and $\varphi = 0.26$ h/km. When an off-ramp queue spillover occurs within main-line section a, the rightmost lane is occupied by queueing vehicles, and the three lanes drop to two lanes; i.e., the number of dropped lanes $\Delta\lambda = 1$. The parameters ξ_v and ξ_ρ in (5) were, respectively, set as 0.05 and 0.13, indicating that an off-ramp queue spillover results in a 5% reduction in free flow speeds and a 13% reduction in critical density. This setting was aligned with the empirical study of [41].

We ran the simulation with no control in the first hour, i.e., 5–6 a.m., to warm up and prepare the background traffic conditions. The remaining hours in the simulation period (6–9 a.m.) were utilized for testing the effectiveness, applicability, and robustness of the proposed strategy.

Effectiveness of the Proposed Control Model

For the comparative analysis, the case study implemented three ramp metering strategies: 1) the classic ALINEA control strategy [18], an effective local feedback control that has been applied in many freeways worldwide; 2) a traditional CRM strategy with no consideration of off-ramp dynamics; and 3) the proposed CRM strategy. All three strategies were compared with the no-control case.

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The control parameters of the proposed strategy were set as follows. The MPC algorithm used 90 time intervals (15 min) for the prediction horizon N_p. The control step size N_c was set as 30 time intervals; i.e., the control actions were updated every 5 min. In addition, the proposed strategy required the minimum metering rates $\mu_i^{\min} = 0.1$ for all on-ramps $i \in \boldsymbol{I}_{\text{ramp}}$.

The traditional CRM strategy was different from the proposed one in the following two respects. First, off-ramp traffic dynamics were not considered in the modeling, constraints, and objective functions of the optimization. The $\phi_{a,t}$ in (4)–(6) was always equal to zero, which means the off-ramp queue spillover and its impact on mainstream traffic flows were not taken into account. The optimization had no constraint on off-ramp queues (constraint 9b). Additionally, the TTS in the objective function did not include the TWT at off-ramps. Therefore, the objective function did not contain the third and the last term of (10). Second, it assumed that the off-ramp flow was a preset and fixed ratio of the mainstream flow. According to historical traffic volumes, the split ratio at off-ramp 1 was set as 0.1, indicating that 10% of the mainstream flows exit at off-ramp 1, and the remaining 90% proceed to the downstream main line. Similarly, the split ratios at off-ramps 2 and 3 were set as 0.4 and 0.3, respectively.

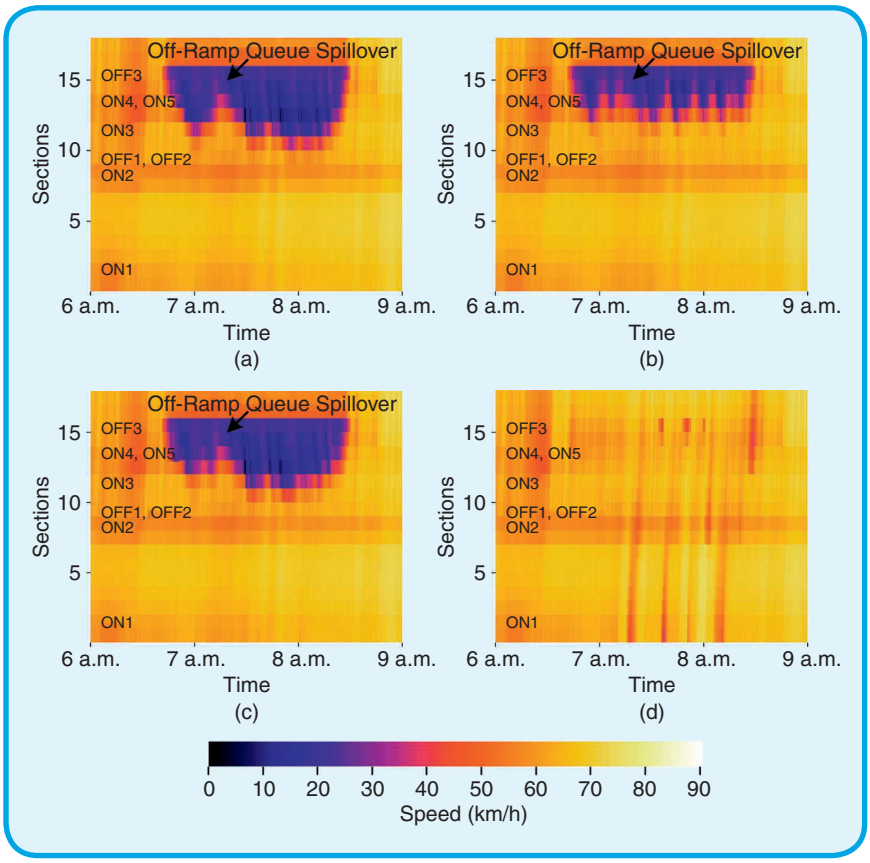


FIG 7 The mainstream speed contour plots of different control strategies: (a) no control, (b), ALINEA, (c) the traditional CRM, and (d) the proposed CRM.

Regarding ALINEA, the metering rates for each on-ramp were updated independently according to a feedback control law. In other words, all five on-ramps used ALINEA control, but there was no coordination between them. A queue override will take place to prevent on-ramp queue spillover. For the equations and parameter settings of ALINEA control, readers can refer to [18].

Figures 7, 8, and 9, respectively, compare the traffic dynamics of the mainstream, on-ramps, and off-ramps when different control strategies are implemented. In particular, Figure 7 demonstrates the speed contours of the mainstream. Figures 8 and 9, respectively, display the relative queue lengths of the on-ramps and off-ramps. The relative queue length is a ratio of the queue length to the storage capacity, ranging from zero to one. The performance indexes of different control strategies are summarized in Table 1, including the TTS of the entire network, the TTT on the mainstream, and the TWT at on-ramps and off-ramps.

The no-control case is characterized by severe mainstream congestion due to off-ramp queue spillover. As shown in Figure 7(a), the mainstream congestion starts near off-ramp 3, spreads upstream, and continues from approximately 6:40 to 8:30 a.m. Meanwhile, it can be seen from Figure 9(a) that the relative queue on off-ramp 3 is larger than one and even up to 1.4, implying that a long vehicle queue spills back onto the main line. Such queue spillover creates the blockage of the rightmost lane on the main line, greatly affecting the efficiency of mainstream traffic. In contrast to the long off-ramp queues, there is no on-ramp queue in the no-control case, as shown in Figure 8(a).

The classic ALINEA control successfully reduces the freeway congestion. Compared to the no-control case, it yields a 2.2% reduction in the TTS of the entire freeway network and a 9.5% reduction in the TTT on the mainstream (see Table 1). The traffic congestion is reduced on the mainstream, as shown in Figure 7(b). However, it fails to prevent off-ramp queue spillover [see Figure 9(b)], and, thus, there is still severe mainstream congestion near off-ramp 3 from approximately 6:40 to 8:30 a.m.

Similar to the case of ALINEA control, the traditional CRM could not prevent off-ramp queue spillover. As can be seen from Figures 7(c) and 9(c), off-ramp 3 is oversaturated

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from 6:40 to 8:30 a.m. Additionally, it is interesting to find that, without consideration of the impact on the mainstream due to the off-ramp queue spillover, the optimization-based MPC (traditional CRM) performs even worse than the local feedback control (ALINEA). The TTS only slightly decreases by 0.5%, and the TTT decreases by 2.6% with respect to the no-control case.

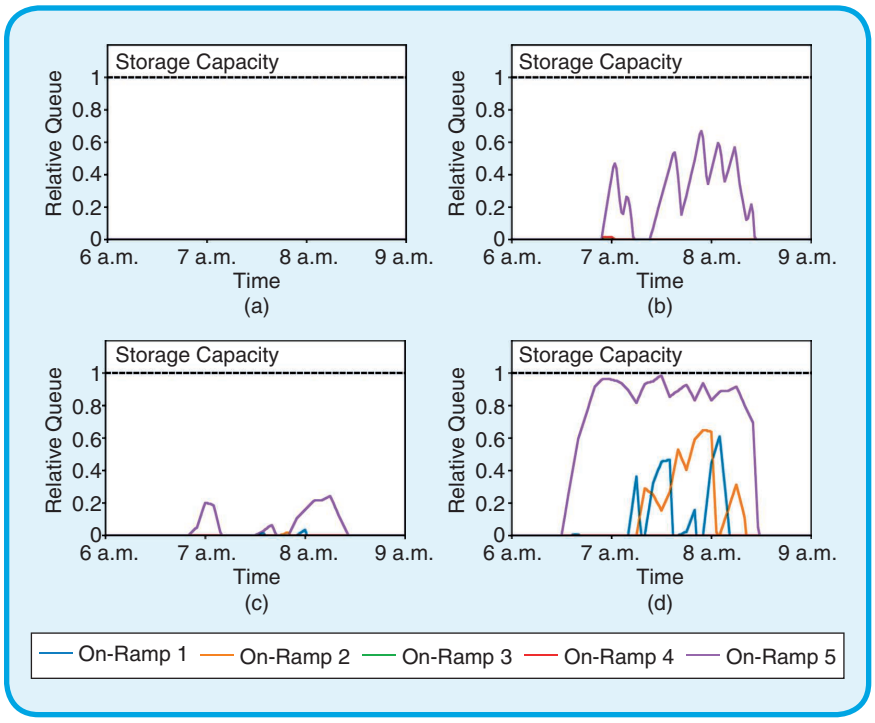


FIG 8 The relative on-ramp queue lengths of different control strategies: (a) no control, (b), ALINEA, (c) the traditional CRM, and (d) the proposed CRM.

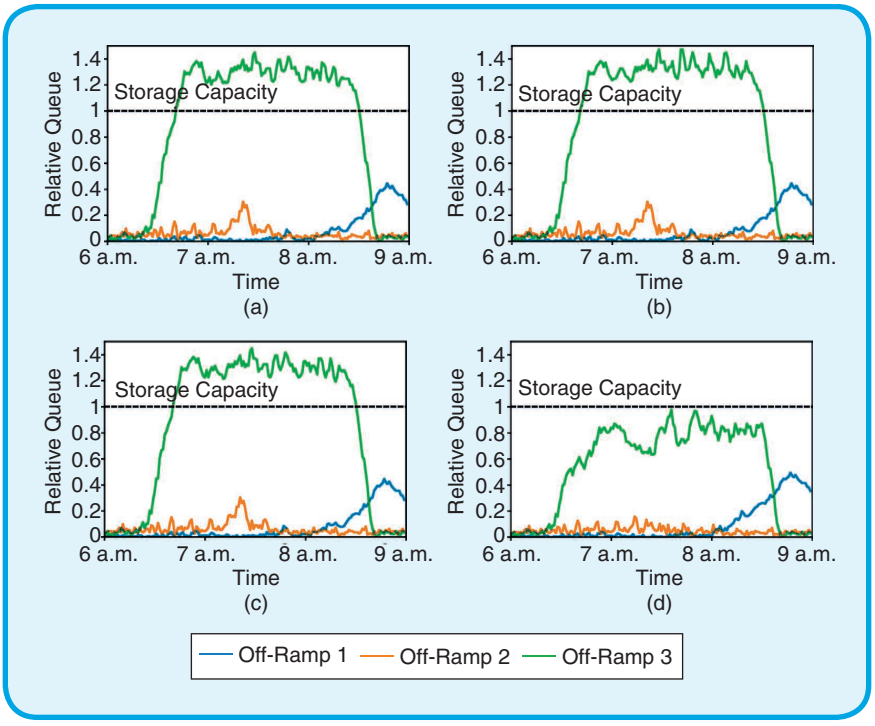


FIG 9 The relative off-ramp queue lengths of different control strategies: (a) no control, (b), ALINEA, (c) the traditional CRM, and (d) the proposed CRM.

In contrast to the traditional CRM, the proposed CRM strategy explicitly considers the off-ramp queue constraint in the optimization to prevent oversaturation [see Figure 9(d)]. Real-time OD information is utilized to predict whether the on-ramp entering vehicles will cause off-ramp queue spillover so that vehicles can be proactively controlled at the on-ramps. By this means, the proposed CRM avoids blockage of the main line due to off-ramp queue spillover. As a result, a high capacity of the main line is maintained, and there is almost no congestion on the mainstream, as shown in Figure 7(d). The TTT on the mainstream is significantly reduced by 33.6% compared to the no-control case, which shows great advantages with respect to ALINEA and the traditional CRM (see Table 1). To some extent, the off-ramp queue spillover is transferred to the on-ramp queues. As can be seen from Figure 8(d) and Table 1, the proposed strategy results in longer on-ramp queues compared to other cases. However, the on-ramp queues do not spill back onto the surface streets. Regarding the overall performance, the proposed strategy achieves an 8.3% reduction in the TTS.

Effectiveness of the Modified Solution Algorithm

In the case study, the optimal control problem was solved by using the regular and modified RPROP algorithms. For both RPROP algorithms, the parameter settings can be found in [45]. Regarding the stopping criterion, the regular RPROP only tests the convergence, while the modified RPROP also verifies whether the convergence value was the minimum of all iterations.

Figure 10 illustrates the convergence of two solution algorithms. It can be seen that the regular RPROP algorithm stops at iteration 3, as marked

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Network Parameters

The parameters of on-ramps, off-ramps, and main-line sections in the case study can be, respectively, found in Tables S1, S2, and S3.

Table S1. On-ramp parameters.

	Number of Lanes λ_i^{ON}	Storage Capacity S_i^{ON} (vehicles)	On-Ramp Capacity r_i^{max} (vehicles/h)
On-ramp 1	2	146	3,600
On-ramp 2	2	176	3,600
On-ramp 3	2	122	3,600
On-ramp 4	2	140	3,600
On-ramp 5	2	190	3,600

Table S2. Off-ramp parameters.

	Number of Lanes λ_j^{OFF}	Storage Capacity S_j^{OFF} (vehicles)	Off-Ramp Capacity e_j^{max} (vehicles/h)
Off-ramp 1	2	155	900
Off-ramp 2	2	112	2,160
Off-ramp 3	2	89	1,260

Table S3. Parameters of the main-line sections.

Main-Line Section Index a	Length L_a (m)	Number of Lanes λ_a	Free Flow Speed v_a^F (km/h)	Critical Density ρ_a^C (vehicles/km/lane)	Jam Density ρ_a^J (vehicles/km/lane)	Capacity q_a^{max} (vehicles/h)	Parameter α_a
1	300	3	83.4	39.6	166.7	5,595	1.75
2	300	3	82.3	37.5	166.7	5,229	1.75
3	300	3	81.8	38.5	166.7	5,334	1.75
4	300	3	82.1	36.7	166.7	5,103	1.75
5	300	3	82.1	36.7	166.7	5,103	1.75
6	300	3	80.6	37.3	166.7	5,043	1.72
7	300	3	80.6	37.3	166.7	5,043	1.72
8	300	3	80	37.8	166.7	4,986	1.67
9	300	3	73.9	40.4	166.7	4,830	1.62
10	300	3	79.4	35.9	166.7	4,683	1.66
11	300	3	80.4	35.4	166.7	4,692	1.67
12	300	3	78.4	39.6	166.7	5,208	1.72
13	300	3	80.7	38.8	166.7	5,217	1.7
14	300	3	80.5	38.7	166.7	5,226	1.72
15	300	3	81.9	37.8	166.7	5,121	1.68
16	300	3	80.9	36	166.7	4,884	1.72
17	300	3	78.9	37.8	166.7	5,001	1.72
18	300	3	80.4	37	166.7	5,040	1.75

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Table 1. Performances of different control strategies.*

Control Strategy	TTS (veh·h)	TTT (veh·h)	TWT^{ON} (veh·h)	TWT^{OFF} (veh·h)
No control	1,634	1,346	0	288
ALINEA	1,599 (–2.2%)	1,218 (–9.5%)	89	292
Traditional CRM	1,626 (–0.5%)	1,311 (–2.6%)	26	289
Proposed CRM	**1,498 (–8.3%)**	**893 (–33.6%)**	**400**	**205**

*Bold values are the performance indexes resulted from the proposed control strategy. Values in parentheses are the percent change as compared to the no-control case.

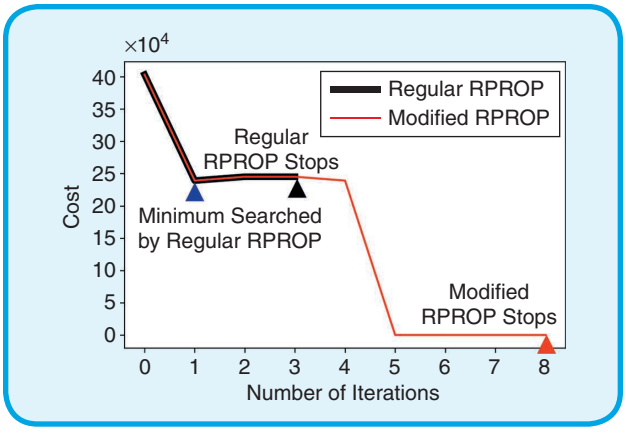


FIG 10 The convergence curves of the different solution algorithms.

by the black symbol. However, the convergence value is larger than the minimum of the previous iterations, as marked by the blue symbol at iteration 1. Therefore, the modified RPROP algorithm rolls back to the control action of iteration 1 and searches this region with smaller increments of the control actions. The modified algorithm then finds control actions to satisfy the off-ramp queue constraints and, thus, enable the penalty terms in (10) become equal to zero, so that the cost criterion J drops dramatically. The algorithm eventually stops at iteration 8, as represented by the red symbol in Figure 10.

The modified RPROP algorithm can converge to control actions that satisfy the off-ramp queue constraints. In contrast, the regular RPROP stops searching at earlier iterations and, thus, fails to prevent off-ramp queue spillover in some time intervals. Therefore, the modified RPROP algorithm generally has better performance indexes compared to the results of the regular RPROP, as shown in Table 2.

Table 2. A comparison of different solution algorithms.*

Solution Algorithms	TTS (veh·h)	TTT (veh·h)	TWT^{ON} (veh·h)	TWT^{OFF} (veh·h)
Regular RPROP	1,549 (–5.3%)	905 (–32.8%)	414	229
Modified RPROP	**1,498 (–8.3%)**	**893 (–33.6%)**	**400**	**205**

*Bold values are the performance indexes resulted from the modified algorithm. Values in parentheses are the percent change as compared to the no-control case.

Sensitivity Analysis

To test the robustness of the proposed strategy, this section analyzes the impacts of the real-time OD information, OD information accuracy, and different control parameter settings.

Impact of Real-Time OD Information

This section analyzes the impact of the real-time OD information on the control performance. The proposed CRM utilizes real-time OD information for traffic inflow and outflow prediction. Some adjustments are made to adapt the proposed CRM to the conditions without real-time OD. Specifically, the split ratios are preset to calculate the off-ramp traffic volumes as the traditional CRM did. The case study compares the CRM without real-time OD and the CRM with real-time OD under three demand levels: 90% (reduce by 10%), 100% (real scenario), and 110% (increase by 10%). Figures 11, 12, and 13, respectively, compare the traffic dynamics of the mainstream, on-ramps, and off-ramps. The corresponding performance indexes are summarized in Table 3.

In general, both CRM strategies have similar traffic performance on the mainstream, on-ramps, and off-ramps (see Figures 11, 12, and 13). Even without real-time OD information, the proposed CRM still manages to prevent off-ramp queue spillover with the appropriate selection of split ratios at off-ramps. The reductions of the TTT on the mainstream become more significant with increasing traffic demands, but the reductions of the TTS regarding to the whole network tend to be less significant as the demand levels increase (see the bold numbers in Table 3).

Impact of Noise OD Information

The case study also tests the sensitivity of the proposed strategy with respect to noise OD information. OD datasets with different levels of noise ranging from 0% to 15% were prepared for the experiments. The levels of noise in the case study denote the mean absolute percent error of the OD matrices for each 5 min [46].

As depicted in Figures 14, 15, and 16, higher levels of noise in the real-time OD information generally lead to similar mainstream congestion and longer waiting times at on-ramps but reduced off-ramp queues. The

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performance indexes are correspondingly summarized in Table 4.

When the real-time OD information is 100% correct (no noise), the proposed strategy manages to prevent off-ramp queue spillover and almost eliminates mainstream congestion by regulating the on-ramp inflows. The overall TTS is reduced by 8.3%.

For the cases with noise OD information, the proposed strategy detects noise destination information during vehicles' traveling and tends to generate stricter metering rates to prevent off-ramp queue spillover. To be specific, if a vehicle sends a destination of off-ramp 1, but it has already passed off-ramp 1, such a destination can be detected as noise information, and a new destination is assigned to this vehicle. To guarantee off-ramp queue spillover prevention, the proposed strategy assigns its destination to the most saturated downstream off-ramp (i.e., off-ramp 3 in this case). Therefore, with higher levels of noise information, the proposed strategy can still prevent off-ramp queue spillover (see Figure 16). In fact, the stricter metering rates reduced the off-ramp queues by sacrificing on-ramp waiting time. As can be seen from Figure 15 and Table 4, on-ramp 5 (the upstream on-ramp nearest to off-ramp 3) is almost saturated from 7 to

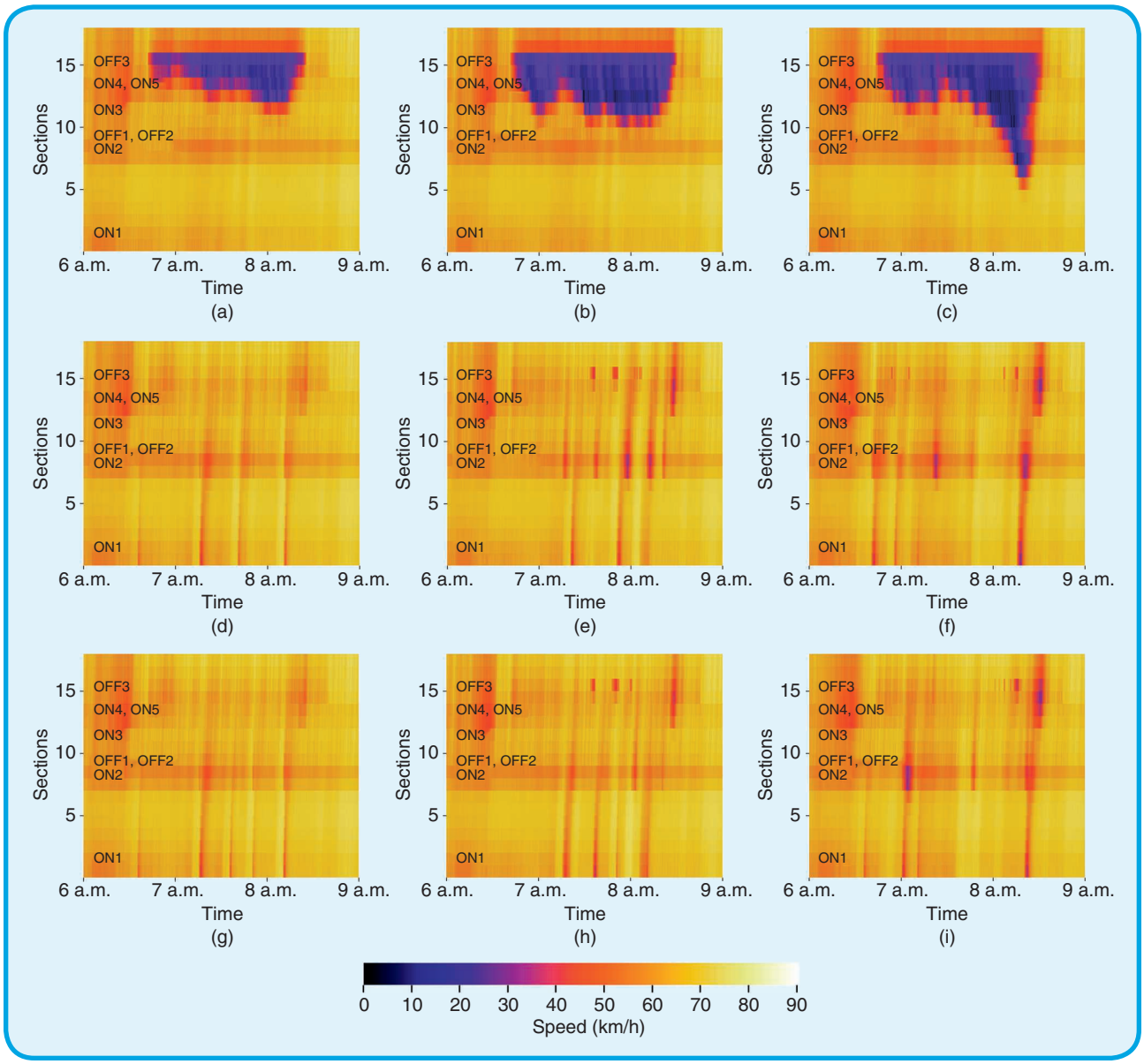


FIG 11 The mainstream speed contour plots of different control strategies under different levels of demand: (a) no control and 90% demand, (b) no control and 100% demand, (c) no control and 110% demand, (d) CRM without real-time OD and 90% demand, (e) CRM without real-time OD and 100% demand, (f) CRM without real-time OD and 110% demand, (g) CRM with real-time OD and 90% demand, (h) CRM with real-time OD and 100% demand, and (i) CRM with real-time OD and 110% demand.

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8:10 a.m. Therefore, with the increasing levels of noise, more vehicles wait at upstream on-ramps 1 and 2 to avoid downstream off-ramp spillover.

In general, when the levels of noise increases, the control performance degrades with respect to the TTS of the entire network. Even though the real-time OD data contain 10% errors, the proposed strategy still outperforms the traditional CRM and ALINEA in terms of the TTS. When the level of noise reaches 15%, the TTS can still be reduced by 2.1%, as shown in Table 4.

Comparison of Different Control Parameter Settings

This section investigates the control performance with different control parameter settings.

Prediction Horizon

To analyze the impact of the prediction horizon N_p on the control performance, this case study adopts 5-, 10-, and 15-min prediction horizons for the same MPC algorithm. The control performance is shown in Figures 17, 18, and 19 and Table 5.

The prediction horizon affects the timing of the ramp metering being active. Using the 5-min prediction horizon as an example, it enables the MPC algorithm to predict the traffic evolution in the next 5 min; correspondingly, the ramp metering can only be activated 5 min before the occurrence of potential congestion. However, 5-min prediction is not enough for preventing off-ramp queue spillover. As can be seen from Figures 17(a) and 19(a), in some time intervals around 7:50 a.m., vehicle queue off-ramp 3 spills back to the mainstream, though the blockage does not continue for long. In contrast, using 10-min and 15-min prediction horizons successfully prevents the off-ramp queue spillover [see Figure 19(b) and (c)]. The prediction horizon should be long enough for all entering vehicles to arrive at downstream off-ramps; otherwise, the waiting time at the off-ramp has not been counted when determining whether to allow a set of vehicles to enter the main line from the on-ramp.

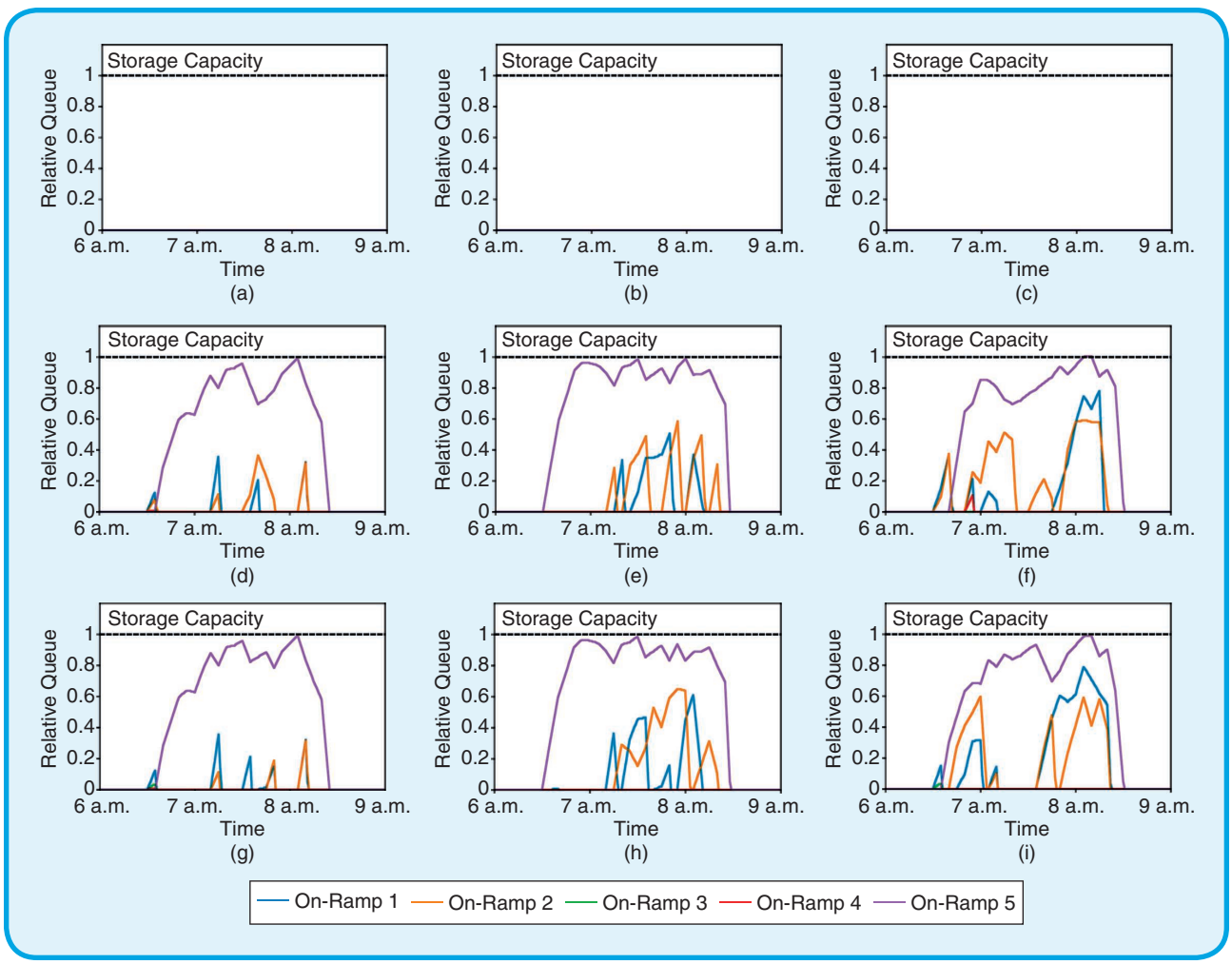


FIG 12 The relative on-ramp queue lengths of different control strategies under different levels of demand: (a) no control and 90% demand, (b) no control and 100% demand, (c) no control and 110% demand, (d) CRM without real-time OD and 90% demand, (e) CRM without real-time OD and 100% demand, (f) CRM without real-time OD and 110% demand, (g) CRM with real-time OD and 90% demand, (h) CRM with real-time OD and 100% demand, and (i) CRM with real-time OD and 110% demand.

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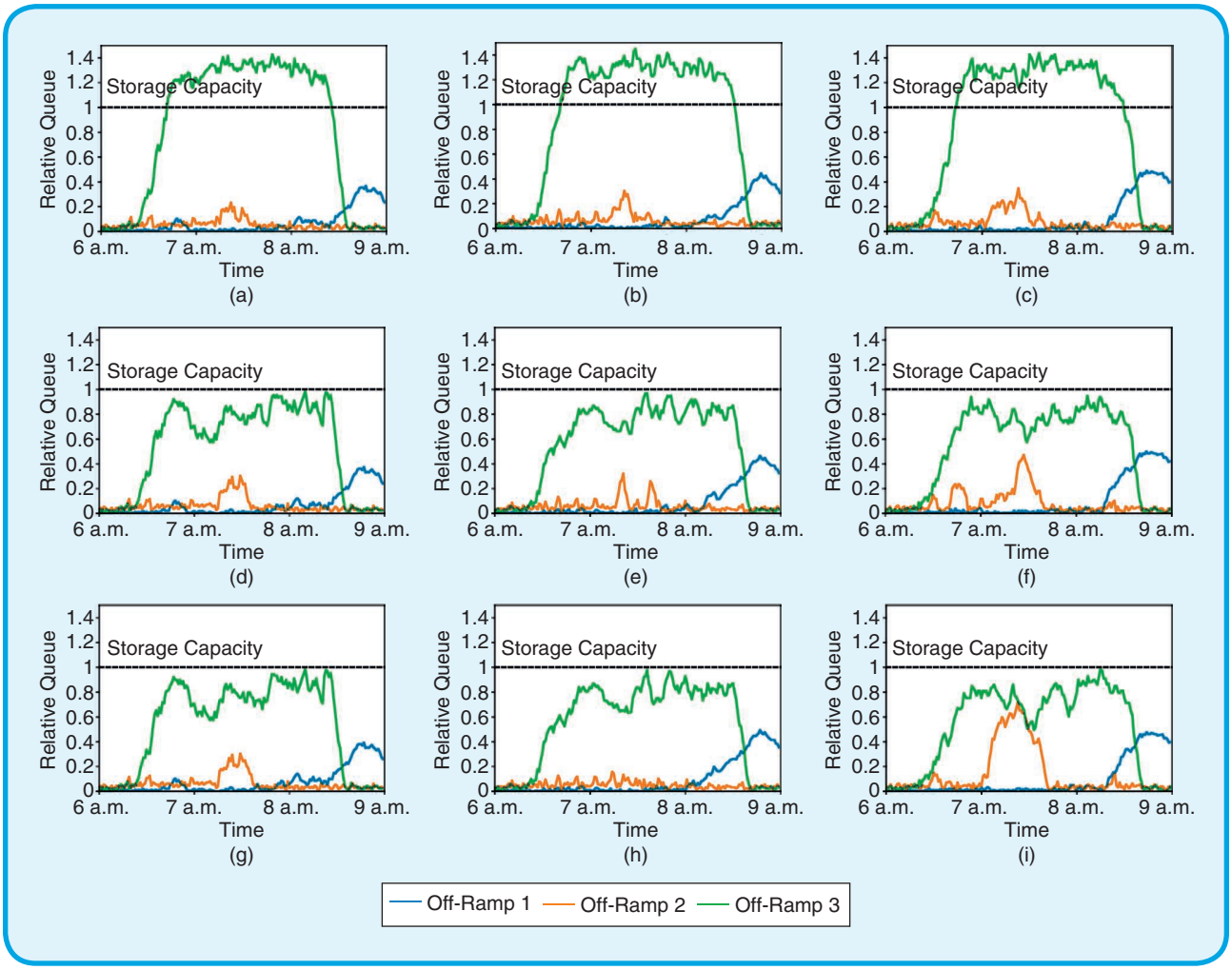


FIG 13 The relative off-ramp queue lengths of different control strategies under different levels of demand: (a) no control and 90% demand, (b) no control and 100% demand, (c) no control and 110% demand, (d) CRM without real-time OD and 90% demand, (e) CRM without real-time OD and 100% demand, (f) CRM without real-time OD and 110% demand, (g) CRM with real-time OD and 90% demand, (h) CRM with real-time OD and 100% demand, and (i) CRM with real-time OD and 110% demand.

Table 3. The performances of different control strategies under different levels of demand.*

Level of Demand	Control Strategy	TTS (veh·h)	TTT (veh·h)	TWT^{ON} (veh·h)	TWT^{OFF} (veh·h)
90%	No control	1,461	1,189	0	272
	CRM without real-time OD	1,358 **(–7.1%)**	885 **(–25.6%)**	274	199
	CRM with real-time OD	1,332 **(–8.9%)**	884 **(–25.7%)**	271	177
100%	No control	1,634	1,346	0	288
	CRM without real-time OD	1,528 **(–6.5%)**	943 **(–29.9%)**	380	205
	CRM with real-time OD	1,498 **(–8.3%)**	893 **(–33.6%)**	400	205
110%	No control	1,691	1,397	0	294
	CRM without real-time OD	1,585 **(–6.3%)**	969 **(–30.7%)**	400	217
	CRM with real-time OD	1,563 **(–7.6%)**	920 **(–34.2%)**	416	228

*Bold values in parentheses are the percent change as compared to the no-control case.

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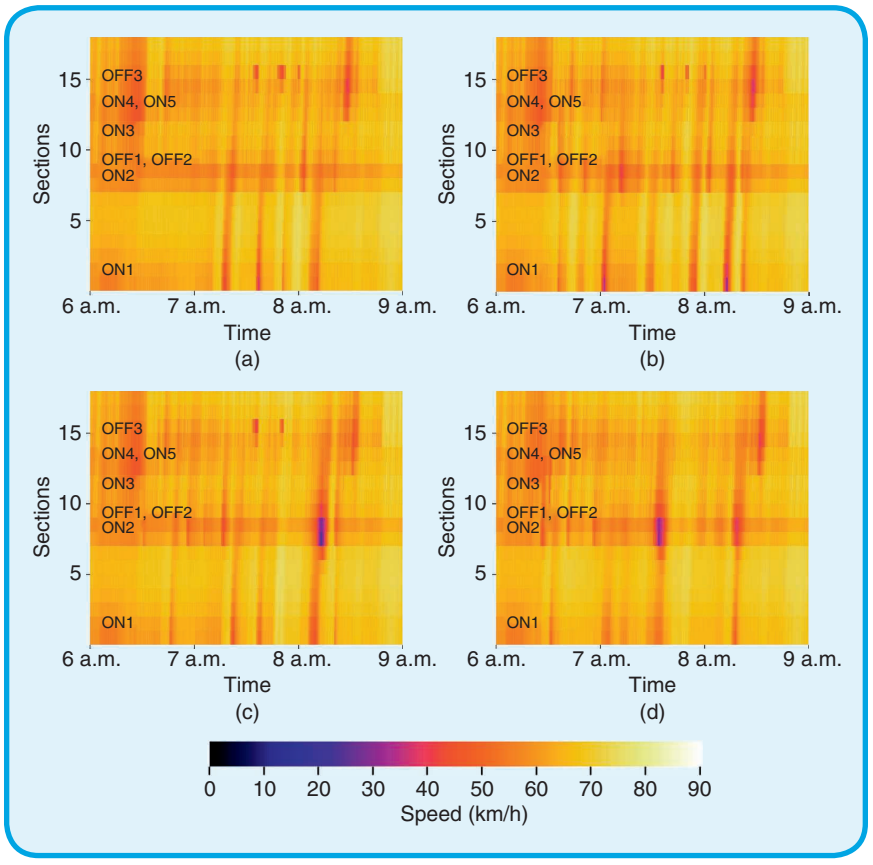


FIG 14 The mainstream speed contour plots of the proposed control strategy under different levels of noise in real-time OD information: (a) no noise, (b) 5% noise, (c) 10% noise, and (d) 15% noise.

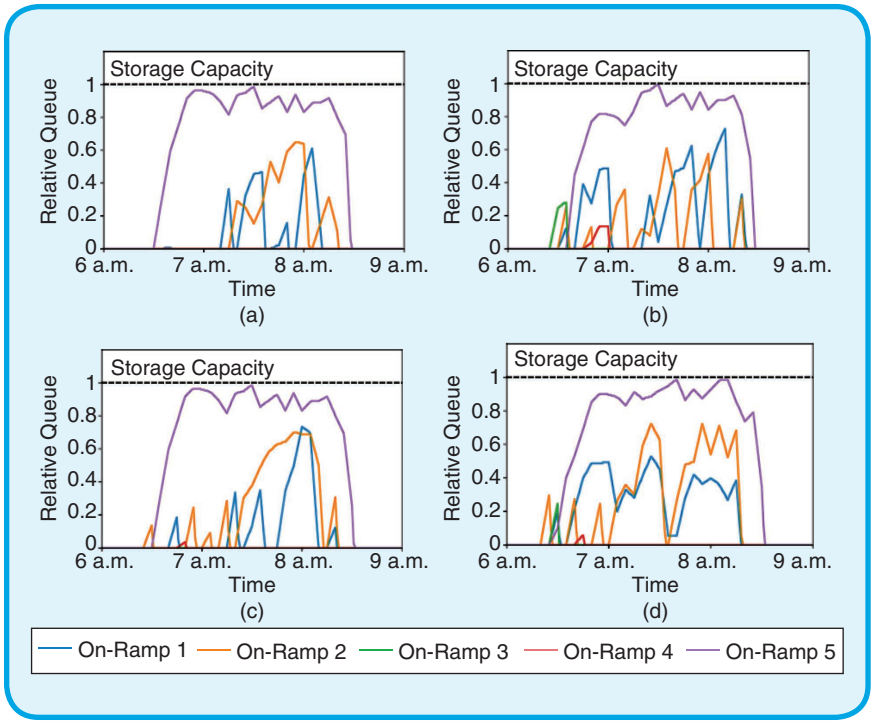


FIG 15 The relative on-ramp queue lengths of the proposed control strategy under different levels of noise in real-time OD information: (a) no noise, (b) 5% noise, (c) 10% noise, and (d) 15% noise.

Additionally, the 5-min prediction horizon causes oscillations in mainstream traffic speeds, on- and off-ramp queues, as can be seen from Figures 17(a), 18(a), and 19(a). It is interesting to find that the 5-min and 10-min prediction horizons result in similar waiting times at off-ramps (see Table 5). However, the queue length at off-ramp 3 tends to be less fluctuated when using 10-min prediction horizons. This is beneficial for preventing off-ramp queue spillover and mainstream blockage. In general, longer prediction horizons lead to reduced TTT and TTS, as can be seen from Table 5.

Control Step Size

The case study also sets different control steps N_c for the same MPC algorithm. The corresponding control performance is shown in Figures 20, 21, and 22 and Table 6. The ramp metering with the 15-min control step size performs the worst with significantly longer waiting time at on-ramps (see Table 6) and obvious traffic jams caused by the entering flows at on-ramp 2 [see Figure 20(c)]. The reason is that 15 min per update is not frequent enough to address the highly dynamic freeway traffic system. Despite that, the proposed control strategy with 15-min control step size can still prevent off-ramp spillover in this case study. When the update frequency increases to 5 min per update, the TTS reduction (in percentage) improves from 2.7% to 8.3%. However, there are less significant differences in the control performance between updating by 5 min and updating every minute. The smaller control step size (larger control update frequency) generally leads to reduced TTS and TWT at on-ramps but similar TTT on the mainstream and TWT at off-ramps.

Conclusion

In this article, a destination-aware CRM strategy is proposed, aiming to prevent off-ramp queue spillover and mainstream congestion

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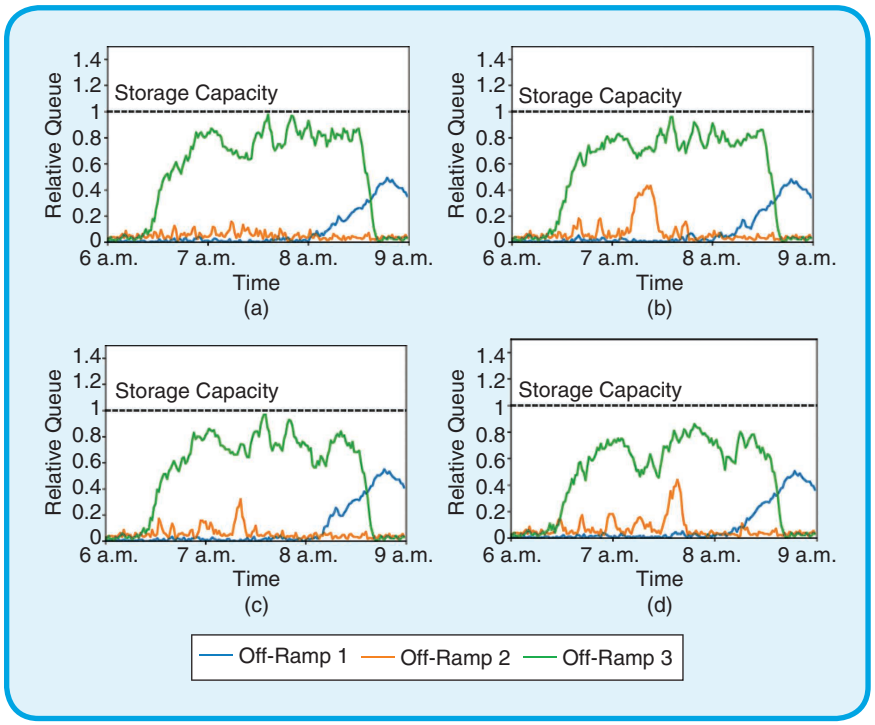


FIG 16 The relative off-ramp queue lengths of the proposed control strategy under different levels of noise in real-time OD information: (a) no noise, (b) 5% noise, (c) 10% noise, and (d) 15% noise.

Table 4. The performance of the proposed control strategy under different levels of noise in real-time OD information.*

Level of Noise	TTS (veh·h)	TTT (veh·h)	TWT^{ON} (veh·h)	TWT^{OFF} (veh·h)
0	1,498 (–8.3%)	893 (–33.6%)	400	205
5%	1,523 (–6.8%)	899 (–33.2%)	414	210
10%	1,545 (–5.5%)	899 (–33.2%)	440	206
15%	1,600 (–2.1%)	898 (–33.3%)	511	191

*Values in parentheses are the percent change as compared to the no-control case.

in urban freeway networks. The proposed strategy explicitly considers the constraints of the off-ramp storage capacity. Such limitations are ubiquitous in high-density urban areas but have long been neglected. The impact on mainstream traffic due to an off-ramp queue spillover is also taken into account in the traffic modeling. An optimization is then designed to minimize the TTS not only on the mainstream and on-ramps but also on the off-ramps. This optimal control problem is solved by a modified RPROP algorithm and embedded in the MPC. In the MPC, the traffic evolution is described from the upstream on-ramp to the main line and, eventually, to the downstream off-ramp so that entering flow can be controlled at the on-ramps if it will potentially cause mainstream congestion or off-ramp queue spillover.

To assess the benefits of the proposed strategy, a comprehensive case study was conducted on an urban freeway network in Shanghai, China, where on-ramps and off-ramps are densely located, and the distribution of OD demand is highly dynamic. The results from the case study show that the proposed CRM strategy reduces the TTS by 8.3% relative to the no-control case, outperforming the traditional CRM and ALINEA control. In addition, the off-ramp queue constraint explicitly

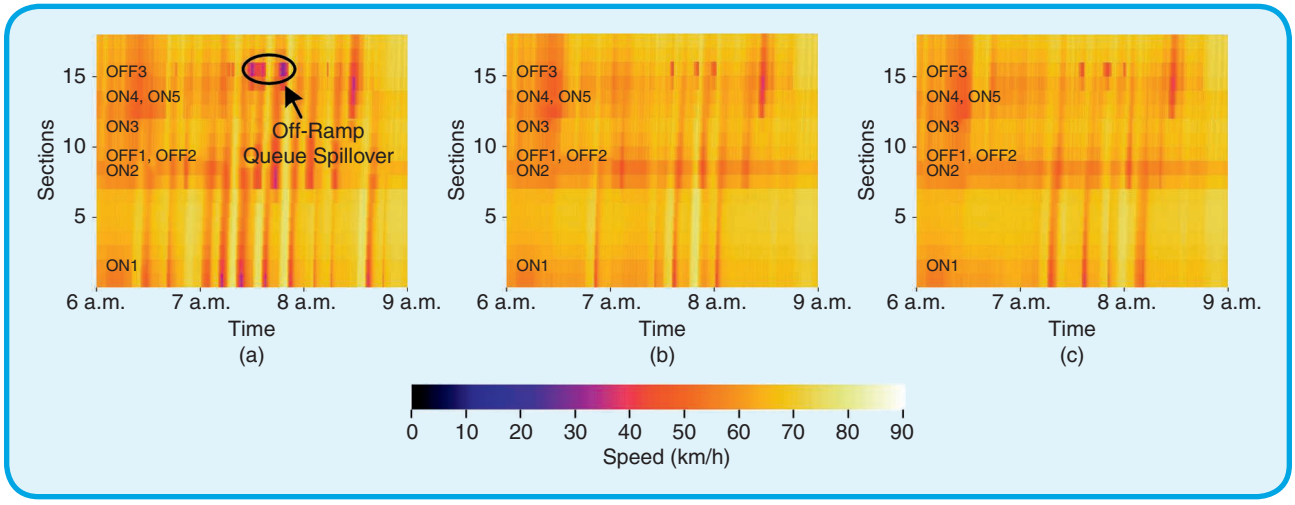


FIG 17 The mainstream speed contour plots of the proposed control strategy with different prediction horizons of (a) 5 min, (b) 10 min, and (c) 15 min.

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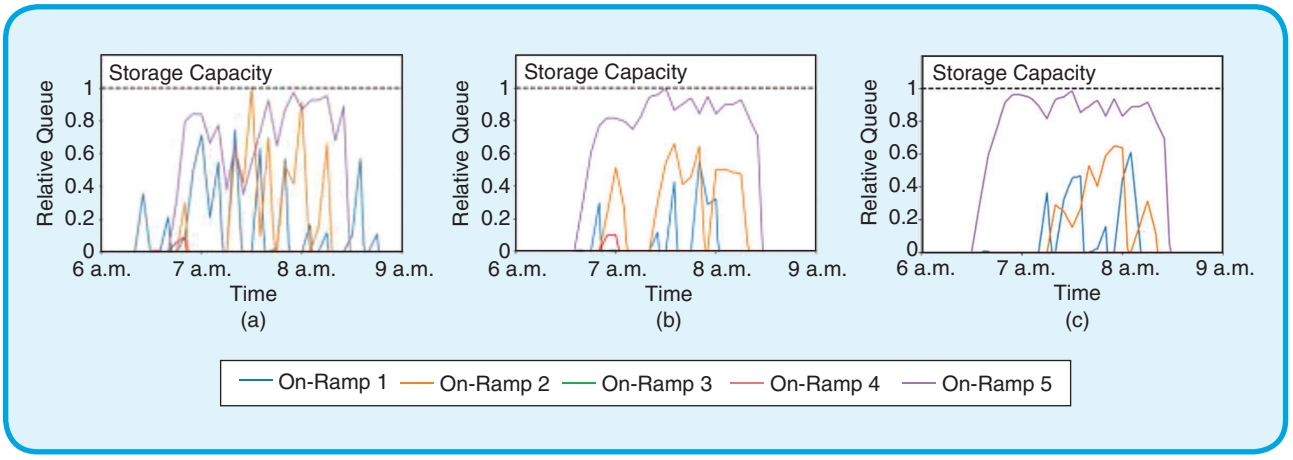


FIG 18 The relative on-ramp queue lengths of the proposed control strategy with different prediction horizons of (a) 5 min, (b) 10 min, and (c) 15 min.

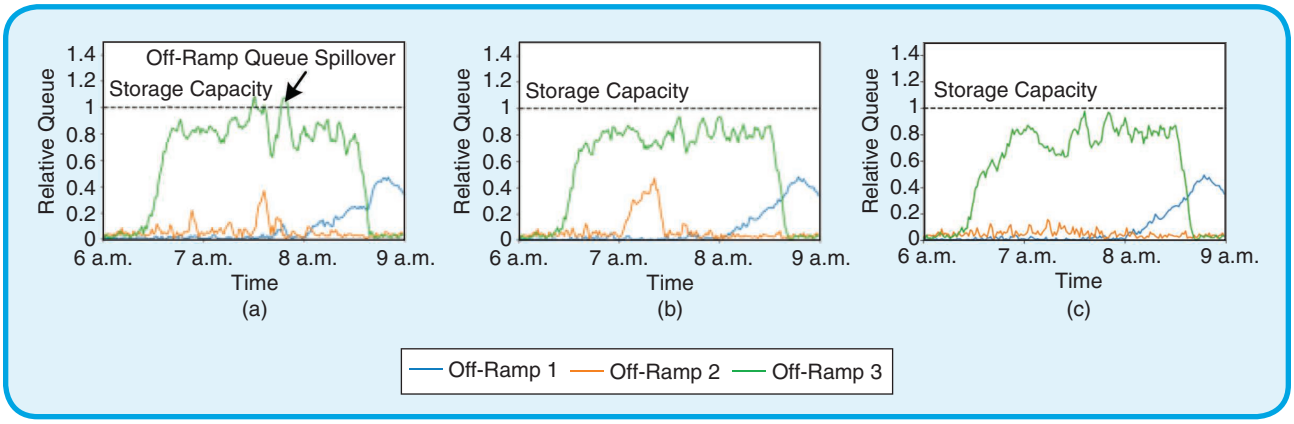


FIG 19 The relative off-ramp queue lengths of the proposed control strategy with different prediction horizons of (a) 5 min, (b) 10 min, and (c) 15 min.

Table 5. The performance of the proposed control strategy with different prediction horizons.*

Prediction Horizon (min)	TTS (veh·h)	TTT (veh·h)	TWT^{ON} (veh·h)	TWT^{OFF} (veh·h)
5	1,562 (–4.4%)	960 (–28.7%)	384	218
10	1,527 (–6.6%)	918 (–31.8%)	391	218
15	1,498 (–8.3%)	893 (–33.6%)	400	205

*Values in parentheses are the percent change as compared to the no-control case.

considered by the proposed strategy can effectively prevent off-ramp queue spillover. When the OD noise is within 10%, the proposed strategy still manages to prevent off-ramp queue spillover and can reduce the TTS by more than 5.5%. With appropriate traffic sensor locations, the OD accuracy can be obtained with less than 10% noise [46], [47], [48]. Even if OD information is not available, the proposed CRM can still prevent off-ramp queue spillover by selecting appropriate split ratios at off-ramps.

The proposed strategy can be realized with traditional ramp signals. With automotive radar and vehicular communication, it can also be used in intelligent transportation systems to control connected and automated vehicles on highways [49], [50], [51]. To apply to most engineering scenarios where only partial OD data are available, enough OD samples are required to reflect the OD distribution of all vehicles. Further investigation is needed to determine the critical OD sampling ratio. Additionally, the destinations and routes of vehicles may be affected by real-time traffic conditions [52]. Therefore, the integration of ramp metering and route guidance with consideration of off-ramp queue constraints is worth investigating. Moreover, computational efficiency should receive attention in future studies to enhance applicability in engineering scenarios.

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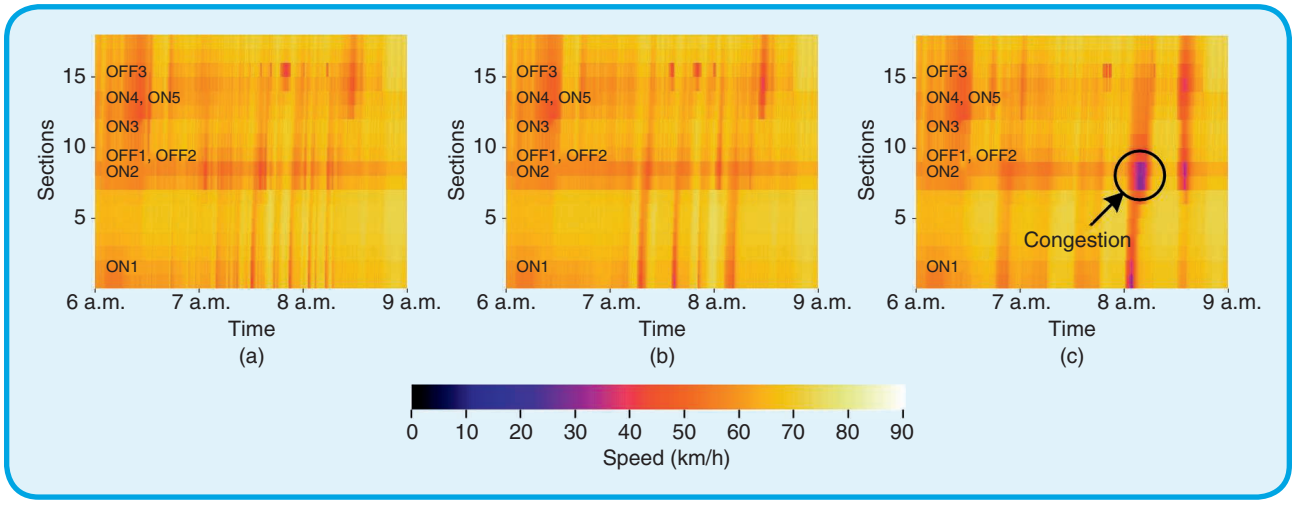


FIG 20 The mainstream speed contour plots of the proposed control strategy with different control step sizes of (a) 1 min, (b) 5 min, and (c) 15 min.

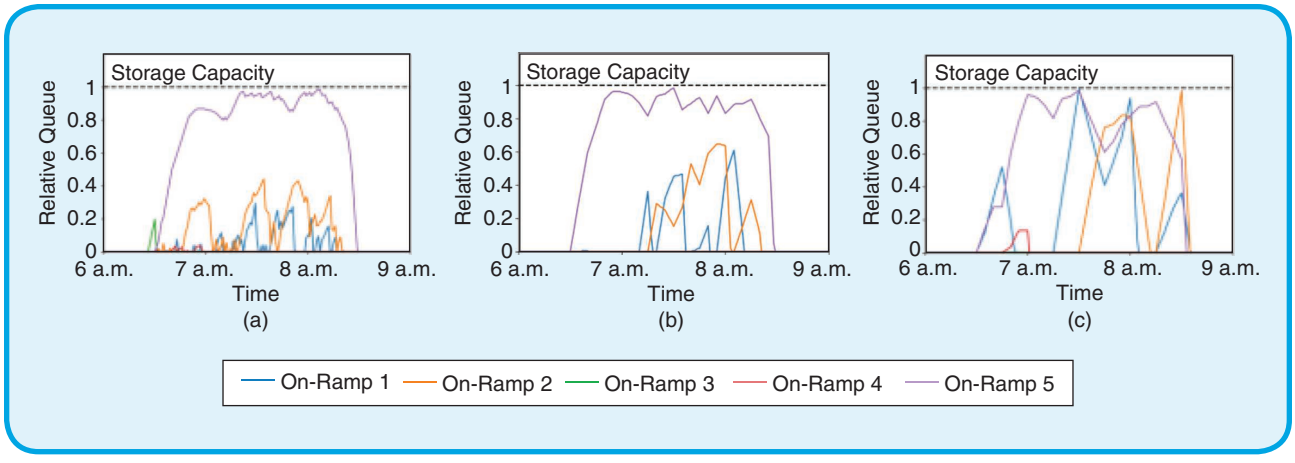


FIG 21 The relative on-ramp queue lengths of the proposed control strategy with different control step sizes of (a) 1 min, (b) 5 min, and (c) 15 min.

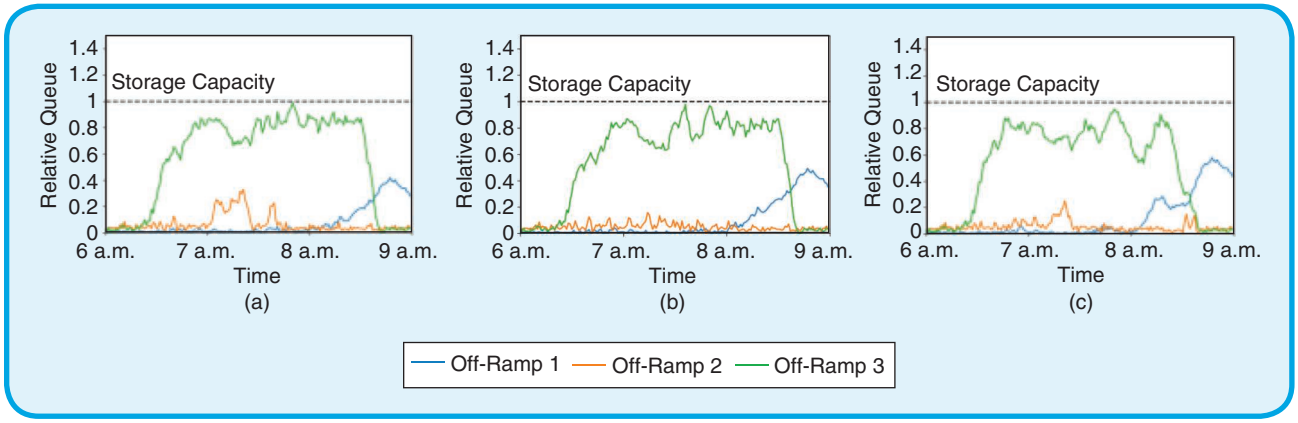


FIG 22 The relative off-ramp queue lengths of the proposed control strategy with different control step sizes of (a) 1 min, (b) 5 min, and (c) 15 min.

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Table 6. The performance of the proposed control strategy with different control step sizes.*

Control Step Size (min)	TTS (veh·h)	TTT (veh·h)	TWT^{ON} (veh·h)	TWT^{OFF} (veh·h)
1	1,472 (–9.9%)	892 (–33.7%)	373	207
5	1,498 (–8.3%)	893 (–33.6%)	400	205
15	1,591 (–2.7%)	908 (–32.6%)	475	207

*Values in parentheses are the percent change as compared to the no-control case.

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